

Answer Sheet

Degree

1) 1000

2) $1000 \times 1000 = 1000000$; A more general answer is: d^k , where d is the number of friends that each of your friends has (presuming that everybody has the same number of friends) and k is the number of paths away from ego (friends of friends; friends of friends of friends, friends of friends of friends of friends, etc.). Thus, the number of friends of friends in this example is $1000^3 = 1000,000,000$.

An more realistic answer to the question of the number of friends of friends (D) that you have (dropping the assumption that everybody has the same number of friends) is given by:

$$D = \sum_{i=1}^K d_i \quad (1)$$

Where K is your total number of friends, $i=1, 2, 3 \dots K$, and d_i is the number friends of the i^{th} person.

Clustering

1) 25

2) $10(10-1)/2$; more general answer, for the total number of undirected dyads (g) among n individuals is:

$$g = \frac{n(n-1)}{2} \quad (2)$$

For undirected ties it is simply $g*2$.

3) For any two contacts a and b in your network the probability that any given contact j already knows any other contact k is given by the sum of dyads that are already realized (the answer to question 1), divided by the total number of possible dyads among your n closest friends. In this example $n=10$. Thus:

$$p(j \leftrightarrow k) = \frac{25}{\frac{1}{2}(10(10-1))} = 0.56 \quad (3)$$

The chances of any two friends not knowing one another is simply one minus this last quantity (0.44).

4) In analogy to step 2 of the Degree questions, the probability of you knowing a friend of a friend would be $p(a \leftrightarrow b)^k$ where k is the number of paths away from ego. Thus $0.56^2=0.31$ is the probability of you knowing a friend of your friend.

5) The number obtained in step 2 of the Degree questions is too large because it assumes zero clustering across different persons. Clustering means that you will share friends of friends, and thus the simple formula 1000^2 "double counts" a substantial number of individuals. The number of persons that are double-counted is given by the probability of you knowing a friend of a friend. Thus, we can correct the answer above by incorporating the answer given in step 4. The expected number of friends of friends that are already your friend is thus: $1000*(0.56^2*1000)=310,000$. Thus, under these assumptions the true expected number of friends of friends should be 690,000.

Dropping the assumption that everybody has the same the same clustering we can then adjust formula (1) above as follows:

$$D = \sum_{i=1}^K (d_i - d_i p_i \rho) \quad (4)$$

Where d_i is the number of friends of your i^{th} friend, p_i is her clustering coefficient, and ρ is ego's clustering coefficient. The product $p_i \rho$ gives the probability that ego knows a friend of the i^{th} contact. Thus, we subtract the expected number of friends of friends (who are not already your friend) corresponding to the i^{th} contact ($d_i p_i \rho$) from the raw number. Notice that when either p_i and $\rho=0$ we get back to (1), conversely when both p_i and $\rho=1$ (all of contact i 's friends are friends of one another and all of your friends are friends of another), then that contact does not contribute any additional "friends of friends" (because they are already your friend).